

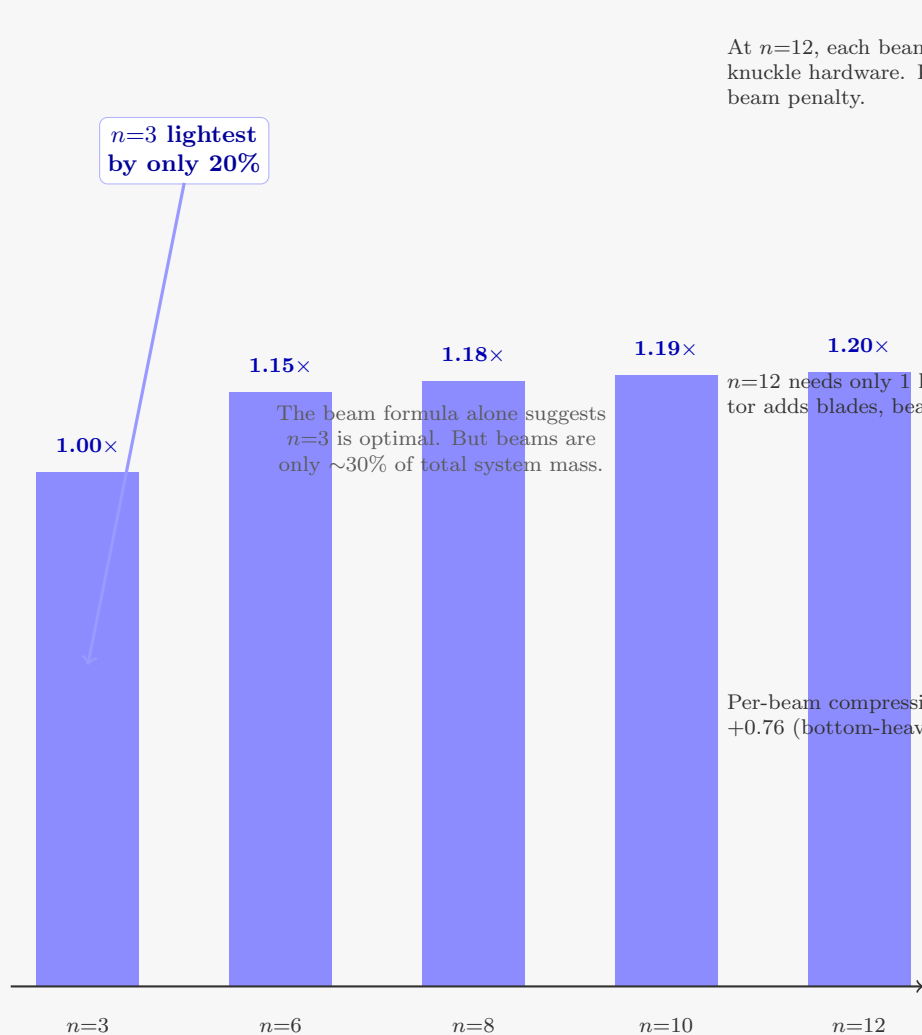
Why $n=12$: Beam Mass vs System Reality

Beam formula predicts only 1.20× penalty — but knuckle coupling & rotor consolidation dominate the system optimum

Beam Mass Model

$m \propto n \cdot \sin(\pi/n)$ — each beam thinner as n rises

Normalised beam mass



System Reality — Three Effects

DE optimiser found $n=12$, 74.17 kg, 58/60 islands converged

1. Knuckle coupling

At $n=12$, each beam is 95 mm vs 118 mm at $n=8$. Thinner beams → dramatically smaller knuckle hardware. Knuckles are ~25% of system mass — this saving dominates the 1.20× beam penalty.

2. Rotor consolidation

$n=12$ needs only 1 hub rotor to distribute torque. $n=8$ needed 3+ expansion rotors. Each rotor adds blades, bearings, and aerodynamic drag. Consolidating to 1 rotor saves ~15–20 kg.

3. Buckling relaxation

Per-beam compression at $n=12$ is $N/12$, vs $N/3$ at $n=3$. Rings can be thinner, β flips from +0.76 (bottom-heavy) to -0.13 (mild top-bias). The entire ring stack gets lighter.

Net: 74.17 kg at $n=12$ — the system optimum

Beam mass penalty: +20%. Knuckle + rotor + ring savings: -40%.